

Kubernetes Orchestration Guide

Introduction

Kubernetes (K8s) is the industry-standard platform for orchestrating containerized applications. It automates deployment, scaling, networking, self-healing, and management of containers across clusters of machines.

Core Topics Covered

- Kubernetes Architecture
- Control Plane Components
- Worker Nodes
- Pods, Deployments, ReplicaSets
- Services and Networking
- ConfigMaps and Secrets
- Persistent Volumes
- Ingress
- Scaling and Autoscaling
- Monitoring and Logging
- Helm
- Security and RBAC
- Cloud Kubernetes (EKS, GKE, AKS)
- Kubernetes for AI and MLOps
- Production Best Practices

This PDF is a generated version of the Kubernetes guide. You can extend it further by adding the remaining chapters from the document.

Summary

Kubernetes provides a powerful platform for running modern cloud-native applications. Understanding pods, deployments, services, storage, security, monitoring, and scaling is essential for DevOps and cloud engineering roles.